A Two-Stage Transfer Learning EfficientNet-B0 Model for Dermatoscopic Image Classification

Nikunj Garg
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
nikunj.garg.ug23@nsut.ac.in

Priyanshu
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
priyanshu-ug23@nsut.ac.in

Saksham Chopra
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
saksham.chopra.ug23@nsut.ac.in

Ayushi
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
ayushi.phd25@nsut.ac.in

Surendra Nagar
Department of Computer Science
Netaji Subhas University of Technology
New Delhi, India
surendra.nagar@nsut.ac.in

***Abstract*—Deep learning for melanoma detection encounters substantial challenges owing to pronounced class imbalance and considerable inter-class visual similarity. This research introduces a lightweight `EfficientNet-B0` framework utilising an innovative two-stage transfer learning technique to improve minority class recognition without elevating computing expenses. The model attains strong generalisation on the DermaMNIST benchmark with the integration of progressive fine-tuning, label smoothing, and class-specific augmentation. The results indicate an accuracy of 87.48%, a macro F1 score of 0.7721, and an AUROC of 0.9792, exceeding the performance of larger ResNet and Vision Transformer baselines. These findings substantiate the effectiveness of small, optimised networks for real-time diagnosis in resource-limited clinical environments.**

***Index Terms*—Transfer Learning, EfficientNet-B0, Skin Lesion Classification, Deep Learning, Medical Imaging, DermaMNIST.**

I. Introduction

Skin cancer is a significant worldwide health issue arising from the unregulated proliferation of atypical cells in the epidermis. The development of this condition is closely linked to extended exposure to ultraviolet (UV) radiation, genetic predisposition, and environmental variables that induce DNA damage in the epidermis. *Melanoma* is the most aggressive form of skin cancer and constitutes a substantial percentage of skin cancer-related fatalities. *Basal cell carcinoma (BCC)* and *Squamous cell carcinoma (SCC)* are two other non-melanoma forms that are also prevalent.

Proper identification of skin abnormalities is essential for decreasing mortality; nonetheless, differentiating malignant from benign lesions poses difficulties due to their analogous clinical presentations, even for seasoned dermatologists. *Dermatoscopic imaging* facilitates noninvasive observation of subsurface skin structures; yet, it necessitates considerable medical knowledge for accurate interpretation and is prone to substantial inter-observer variability.

The application of advances in computer vision and deep learning research has significantly enhanced capabilities in medical image analysis, including dermatological diagnosis. *Convolutional Neural Networks (CNNs)* and attention-based architectures can automatically extract relevant visual information from dermatoscopic images, thereby supporting clinicians in early detection. Some of the well-established models such as *ResNet*, *DenseNet*, *Inception*, and *U-Net* have produced outstanding success in model lesion segmentation and classification [22].

Nevertheless, these systems are computationally intensive and require large amounts of labeled data and high-performance hardware to provide optimal performance. This precludes their use in hospital environments and is therefore difficult to translate into clinical usage, especially in resource-poor or mobile diagnostic environments.

The introduction of the `EfficientNet` family of networks provides a promising trade-off between accuracy and computational efficiency by means of compound scaling of network depth, width, and resolution. Recent research has further explored both CNN and ViT architectures on dermatology benchmark datasets such as HAM10000 [23] and DermaMNIST [19], [21], and shown promising results [2], [1], [3].

However, persistent challenges such as dataset imbalance, low sample diversity, and limited generalization across lesion classes continue to reduce model robustness. In **DermaMNIST**, for example, the images of melanocytic nevi dominate the dataset, while there is poor recognition of the classifiers on the minority lesion classes such as dermatofibroma and vascular, adversely affecting the reliability of diagnosis.

To overcome these problems, the present study aims to develop a compact but effective dermatological lesion classification architecture based on the `EfficientNet-B0` backbone, while employing a **two-stage transfer learning strat-**

egy with fine-tuning of the pre-trained weights on the **DermaMNIST** dataset, while utilizing regularization techniques such as *label smoothing* and *targeted data augmentation*. An *adaptive learning rate schedule* is also applied for optimizing convergence stability, generalization accuracy, and minority class recognition. Diagnostic accuracy should therefore be achieved with minimum computational overhead, so that the architecture could be suited for real-time dermatological applications in resource-poor environments.

The key contributions of this research are as follows. First, a custom EfficientNet-B0 pipeline was developed for DermaMNIST, yielding superior accuracy and parameter efficiency compared to standard CNN and ViT baselines. Second, the model was improved by incorporating label smoothing and adaptive learning rate scheduling, which stabilized training and enhanced resilience to class imbalance. Finally, extensive ablation studies and benchmarking demonstrate that the proposed approach achieves state-of-the-art performance on the DermaMNIST dataset, validating its potential for real-time clinical application

II. Literature Review

Deep learning has significantly transformed dermatological image analysis over the past decade. In the past, methods mostly used hand-made descriptors like colour and texture features, which were then processed by standard machine learning classifiers. The development of Convolutional Neural Networks (CNNs) made it possible to extract features from image data from start to finish, which greatly improved automated skin cancer detection systems. The field's rapid advancements have been summarized in several recent reviews and surveys [13], [14].

Deep learning applied to the analysis of skin lesions has presented a number of potential and obstacles, according to recent studies on dermatological picture categorisation. [1] conducted a detailed evaluation of the DermaMNIST and Fitzpatrick17k datasets and discovered significant issues such low image quality and class imbalance that hindered classification models from consistently generalising. Their study made clear how important standardised assessment and preprocessing methods are for trustworthy benchmarking in dermatological imaging.

Comparative studies of different model architectures have further demonstrated the utility of different techniques for medical imaging. Studies evaluating Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) [2] on several benchmarks, including DermaMNIST, CNNs typically outperform ViTs on smaller datasets due to their intrinsic inductive biases and parameter efficiency. They also demonstrated how hybrid CNN-ViT models might use complementary features to improve classification performance.

Several studies have shown that data-centric techniques are effective. CNN-based classifiers for pigmented lesion diagnosis were trained using the DermaMNIST dataset by [3], demonstrating that sophisticated data augmentation and fine-tuning techniques can greatly increase accuracy, especially in low-data situations. This has led to the development of various effective DCNN [15], hybrid [5], [7], dual-branch [10], and other novel frameworks [4].

The advantages of model variety in improving diagnostic accuracy were also highlighted by [11], who suggested an ensemble framework integrating DenseNet and EfficientNet variants. This framework outperformed individual networks in detecting malignant lesions with modest morphological differences. Other strategies include using segmentation [16] or localization [6] to improve classification focus.

For dermatological analysis that is computationally efficient, transfer learning is still a common approach. For the purpose of detecting melanoma, [12] used EfficientNet-B3 with pretrained ImageNet weights, attaining high classification accuracy while preserving real-time inference capabilities appropriate for clinical deployment. Simultaneously, [17] investigated Vision Transformers (ViTs) for melanoma detection and discovered that although ViTs provide interpretability based on attention, their effectiveness is highly dependent on huge, high-quality datasets.

Enhancing diagnostic reliability has also been demonstrated to be possible through the incorporation of attention mechanisms. By highlighting clinically significant areas in dermatoscopic pictures, [18] developed an attention-guided CNN that increased the sensitivity for detecting malig-nant lesions. The ability of lightweight designs to provide high performance without incurring significant computing costs is further supported by benchmarking tests. In evaluations of more than twenty deep learning systems by [20] and other benchmarking studies [8], it was shown that models such as EfficientNet-B0, MobileNetV2, and other custom CNNs [9] can provide competitive accuracy while still being useful for real-time medical applications.

All things considered, recent studies demonstrate the advantages and disadvantages of contemporary deep learning in dermatology. Despite the high classification accuracy of CNNs and transformer-based designs, their clinical application is frequently limited by their computing requirements. This encourages the employment of lightweight, optimised models, such `EfficientNet-B0`, which can provide a resource-efficient, clinically implementable solution for dermatological image analysis while achieving state-of-the-art performance on **DermaMNIST**.

III. Methodology

The EfficientNet-B0 architecture, which is lightweight and designed for implementation in clinical settings with limited resources, is used in the suggested categorisation framework. The thorough process, which is depicted in Fig. 1, combines a two-stage transfer learning approach with a strong data augmentation pipeline. This method maintains computational economy while optimising diagnostic precision on the DermaMNIST benchmark by combining deep feature extraction with a bespoke classifier head.

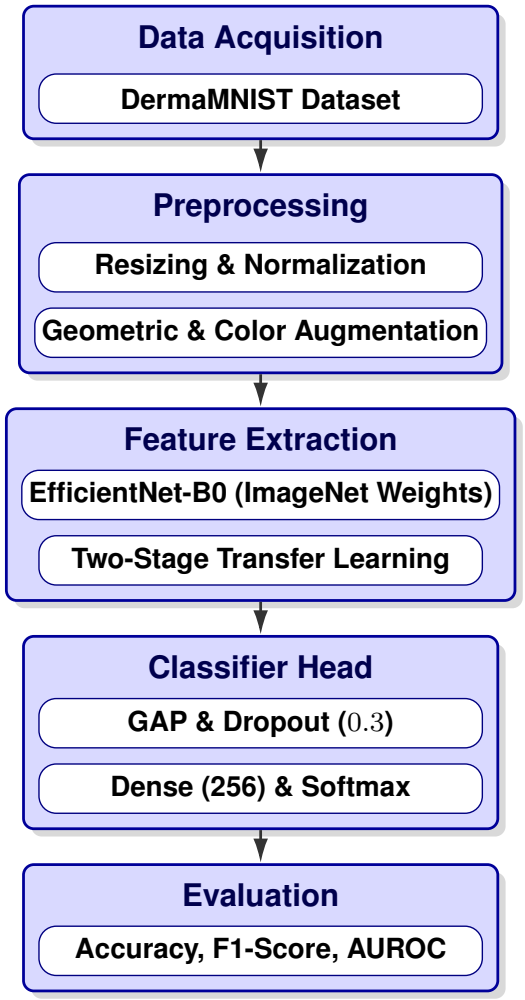

Fig. 1. **Proposed methodological pipeline.** The workflow details data acquisition, geometric preprocessing, and feature extraction via **EfficientNet-B0**. The pipeline utilizes a two-stage transfer learning strategy and concludes with a custom classifier head optimized for lesion diagnosis [19].

A. Dataset Overview

The research experiments were conducted using the DermaMNIST dataset [19], [21], which was created from the well-known HAM10000 collection [23] of dermoscopic pictures. The DermaMNIST dataset contains 10,015 images divided into seven diagnostic classes: *dermatofibroma, benign keratosis-like lesions, basal cell carcinoma, actinic keratoses, vascular lesions, melanocytic nevi* and *melanoma*. Each image was displayed in normalised RGB format after being shrunk to 28×28 pixels. The dataset consists of 7,007 training samples, 1,003 validation samples, and 2,005 testing samples.

There is a clear class disparity, with benign lesions making up the majority of the distribution. The proposed method uses loss control techniques and targeted augmentation during training to solve this problem. The difference is seen in Fig. 2, which shows that Class 5 (*melanocytic nevi*) makes up the majority of the samples. Illustrative cases from each diagnostic group are shown in Fig. 3.

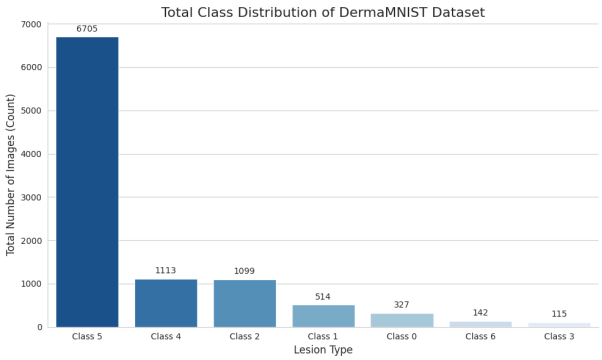

Fig. 2. **DermaMNIST class distribution** [19]. The chart illustrates the severe imbalance between the dominant **Class 5 (melanocytic nevi)** and minority classes, highlighting the need for the proposed class-specific augmentation strategy.

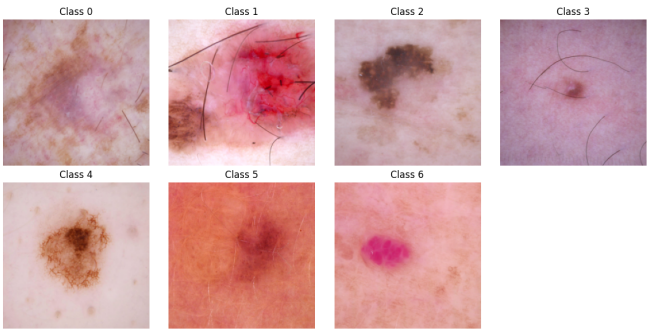

Fig. 3. **Representative dermoscopic samples.** Images from the DermaMNIST dataset [19] displaying **high inter-class visual similarity** and intra-class variation, which pose significant challenges for standard classification models.

B. Preprocessing and Augmentation

The mean and variance determined from the training subset were used to standardise all pictures and normalise them to an intensity range of [0,1]. A thorough augmentation pipeline was used to address data imbalance and enhance generalisation.

The augmentations included zoom variations of up to 10%, rotations between 15° and 30°, random flips in both the horizontal and vertical directions, and colour jittering to change saturation, contrast, and brightness. By reducing overfitting and increasing intra-class variability, these augmentations helped the model better adjust to changes in skin tone, texture, and illumination that occur in the real world during inference.

C. Model Architecture

The `EfficientNet-B0` design, illustrated in Fig. 1, was selected for its balanced trade-off between depth, width, and resolution, achieving excellent accuracy with a comparatively limited number of parameters. ImageNet pre-trained weights were used to initialize the network, and specific architectural adjustments were made to adapt it for seven-class classification.

First, spatial information is compacted efficiently using a Global Average Pooling (GAP) layer, which transforms feature maps into a summarized representation. This is followed by a fully connected layer of 256 ReLU-activated neurons, enabling the model to learn task-oriented representations. To enhance generalization and limit neuron co-adaptation, a Dropout layer ($rate = 0.3$) is introduced. Finally, a Softmax-activated Dense layer with seven output nodes is employed at the network's termination to generate class probabilities across the diagnostic categories. This setup supports effective deployment on moderate hardware without compromising discriminative accuracy due to its expressive yet lightweight design.

D. Two-Stage Transfer Learning Strategy

The **two-stage transfer learning framework**, which gradually adapts previously learnt characteristics to the dermatological imaging domain while reducing catastrophic forgetting, is a key contribution of this study.

1) Stage 1 Feature Extraction: In the first stage, the convolutional base of `EfficientNet-B0` was frozen, and only the classifier head was trained for 15 epochs.

The mapping of general ImageNet characteristics to lesion classifications unique to dermatology is the main goal of this step. General visual representations are preserved and early learning is stabilised when the base network is frozen. The Adam optimiser was used for training, with a learning rate of 1×10^{-3} and a weight decay of 1×10^{-5}.

2) Stage 2 Domain Adaptation and Fine-Tuning: After the classifier stabilised, all layers were unfrozen for joint optimisation on a lowered learning rate of 1×10^{-4} for another 25 epochs. Lower-level filters can adjust to dermatoscopic textures and pigmentation patterns due to this gradual unfreezing, which also guarantees consistent convergence.

ReduceLROnPlateau and early stopping were used with a five-epoch patience to fine-tune the training dynamics. Categorical Cross-Entropy with Label Smoothing ($\epsilon = 0.1$) was used to train the model, which enhances generalisation to unseen lesion types and discourages overconfident predictions.

Compared to single-phase fine-tuning, our two-stage optimisation approach achieves faster convergence, better minority-class performance, and less overfitting by striking a compromise between transfer efficiency and domain adaptability.

E. Evaluation Metrics

ROC-AUC, F1-Score, and Accuracy measures were used to evaluate the model's performance. All provided results represent the average of five-fold cross-validation, ensuring statistical robustness and limiting the influence of dataset partitioning on overall model performance.

IV. Results and Analysis

A. Quantitative Comparison

Table I summarizes the performance of the proposed framework in comparison with other established architectures. The `EfficientNet-B0` model earns the highest F1-score and overall accuracy among all assessed baselines.

TABLE I
PERFORMANCE COMPARISON OF EFFICIENTNET-B0 AND BASELINE MODELS.

Model	Acc.	F1	AUROC
EffNet-B0	**0.8748**	**0.7721**	**0.9792**
EffNetV2-L	0.8490	0.7460	0.9768
EffNetV2-M	0.8401	0.7385	0.9768
ResNet-18	0.8470	0.7450	0.9530
ResNet-50	0.8470	0.7420	0.9480
ViT-Small	0.8156	0.7160	0.9400
ViT-Base	0.8046	0.7050	0.9350

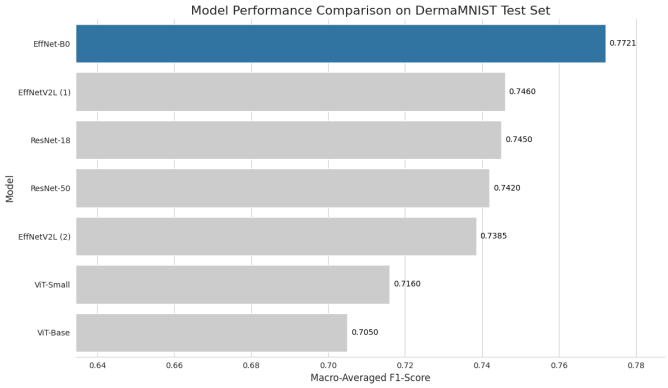

Fig. 4. **Model performance comparison.** A comparison of **Macro-Averaged F1-Scores** shows the proposed EfficientNet-B0 (0.7721) outperforming larger ResNet and Vision Transformer baselines, demonstrating superior efficiency.

The macro-averaged F1-scores across competing models are shown in Fig. 4 to give a visual summary, highlighting the improved performance of our suggested method.

B. Training Dynamics

The trends in training and validation accuracy for both training phases are shown in Fig. 5. Throughout the learning process, the curves show a consistent increase in accuracy without any indications of instability or overfitting, suggesting stable optimisation and efficient generalisation.

C. Confusion Matrix

The confusion matrix in Fig. 6 shows the prediction distribution across the seven lesion types to further evaluate class-wise performance. Although there are some small classification errors across superficially comparable lesion types, especially within the *melanocytic nevi* group, the model consistently performs well across the majority of classes. This behaviour is consistent with the visual overlap seen in some dermatological disorders and the dataset's underlying class imbalance.

D. Ablation Study

To assess the contribution of each enhancement included in the suggested framework, an ablation study was carried out (Table II). To evaluate each modification's individual and combined effects on performance, *data augmentation*, *label*

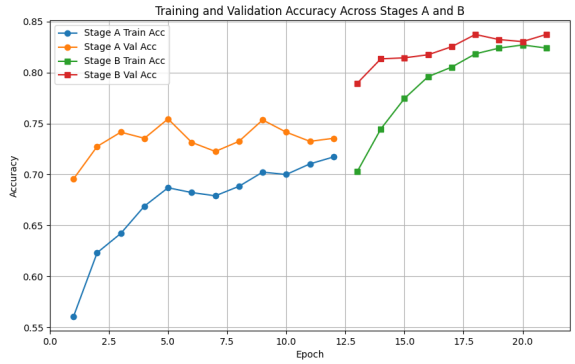

Fig. 5. **Training dynamics.** Accuracy curves across the **two-stage learning process**. Stage 1 (frozen backbone) establishes stable features, while Stage 2 (unfrozen fine-tuning) yields a sharp performance increase without overfitting.

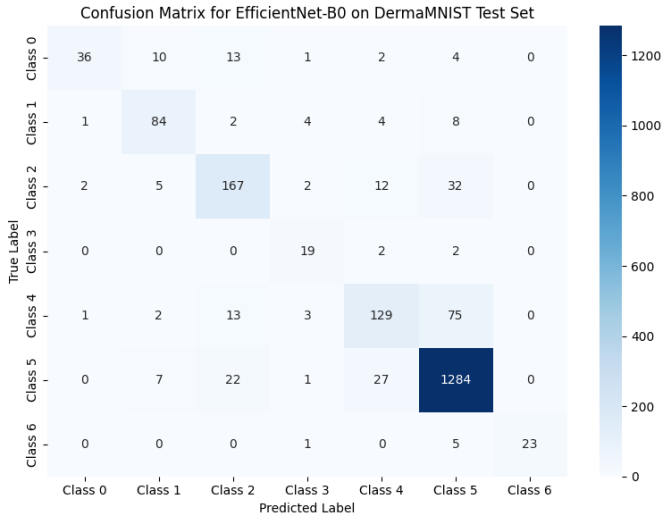

Fig. 6. **Confusion matrix.** The **strong diagonal density** confirms high classification accuracy across most classes. Minimal confusion remains between visually similar categories like melanocytic nevi (Class 5) and benign keratosis (Class 2).

smoothing, and *full fine-tuning* were introduced sequentially to a baseline model.

Fig. 7 ablation results attest to the fact that every improvement raises the overall performance of the model. The biggest benefit comes from data augmentation, which improves generalization to minority groups and reduces class imbalance. Label smoothing and fine-tuning further enhance the learning process by lowering prediction overconfidence and promoting stable convergence. On the **DermaMNIST** benchmark [19], the final model outperformed previously published CNN and Vision Transformer baselines with an **accuracy of 87.48%**, a macro F1-score of **0.7721**, and an **AUROC of 0.9792**.

TABLE II
IMPACT OF AUGMENTATION AND FINE-TUNING STRATEGIES.

Configuration	**Acc. (%)**	**Gain**
Baseline	82.1	–
+ Augmentation	84.5	+2.4
+ Label Smoothing	86.1	+1.6
+ Full Fine-Tuning	**87.4**	+1.3

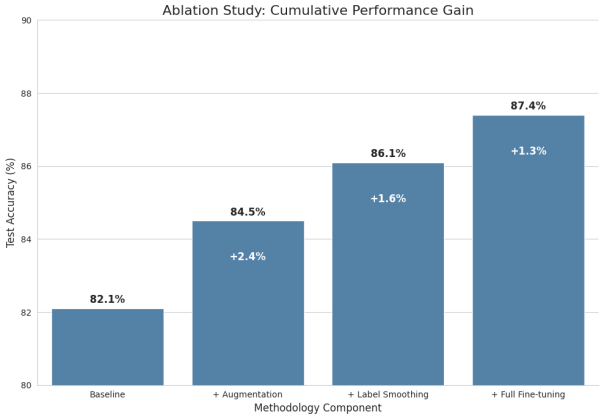

Fig. 7. **Ablation study visualization.** The bar chart depicts the **cumulative accuracy gains** from each module. Data augmentation and the full fine-tuning stage provide the most significant contributions to the final 87.48% accuracy.

V. CLINICAL RELEVANCE AND FUTURE WORK

This study has significant clinical implications. Since benign and malignant lesions can appear quite similar, accurate diagnosis remains challenging, even though early detection of skin cancer, particularly melanoma, improves treatment outcomes. The proposed `EfficientNet-B0`-based framework provides dermatologists with a fast, straightforward, and accurate diagnostic tool to assist in the initial screening and triage of skin lesions.

The model may be used on portable devices like cellphones and handheld dermatoscopes because it is lightweight and only has 5.3 million parameters. Because of this characteristic, the device is particularly useful for teledermatology applications in areas with low resources and limited access to specialists. This strategy may shorten patient wait times and increase diagnostic precision when incorporated into clinical decision-support systems.

Despite its benefits, there are also a lot of difficulties. When employed, the **DermaMNIST** dataset may cause a domain shift since, despite its richness, it does not fairly represent the variety of real-world clinical circumstances or patient demographics, which could result in a domain shift when it is used. Furthermore, interpretability and congruence with clinical reasoning are limited when contextual information, such as patient age, medical history, or lesion development, is missing.

Three main areas will be the focus of future research in order to close the gap between clinical integration and research. In order to increase diagnosis reliability, multimodal

learning will first be investigated by fusing clinical information with dermoscopic pictures. Second, to guarantee consistent performance across different imaging equipment and patient demographics, domain adaptation techniques will be used. Third, in addition to data-efficient learning methodologies to take use of large volumes of unlabelled dermatological data, model explainability will be improved by attention-based visualisation approaches such as Grad-CAM to promote clinician trust.

These improvements seek to bridge the gap between machine learning research and practical dermatological applications by strengthening the framework's adherence to real clinical integration.

VI. Conclusion

This study introduces an optimised `EfficientNet-B0` framework for skin lesion classification, which obtained a macro-averaged F1-score of 0.7721, an AUROC of 0.9792, and an accuracy of 87.48% on the DermaMNIST dataset. The suggested two-stage fine-tuning technique, supported by balanced augmentation and label smoothing, was essential for sustained convergence and enhanced minority class recognition.

The findings demonstrate that appropriately scaled CNN architectures may match or outperform deeper networks like ResNet-50 and Vision Transformers while keeping low computational costs, making them suitable for real-time clinical applications. In order to assess performance in real-world diagnostic procedures, future research will focus on improving model interpretability, cross-domain resilience, and clinical validation trials.

Acknowledgment

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References

[1] K. Abhishek, A. Jain, and G. Hamarneh, "Investigating the quality of DermaMNIST and Fitzpatrick17k dermatological image datasets," *Sci. Data*, vol. 12, p. 82, 2025, doi: 10.1038/s41597-025-04382-5.

[2] A. Amangeldi, A. Taigonyrov, M. H. Jawad, and C. E. Mbonu, "CNN and ViT efficiency study on Tiny ImageNet and DermaMNIST datasets," *arXiv preprint arXiv:2505.08259*, 2025.

[3] N. Kadric, A. Kadribasic, and M. Kapor, "Improving diagnostic accuracy of pigmented skin lesions with CNNs: A case study with DermaMNIST dataset," *arXiv preprint arXiv:2507.12961*, 2025.

[4] A. S. Al-Waisy *et al.*, "A deep learning framework for automated early diagnosis and classification of skin cancers (Skin-DeepNet)," *Sci. Rep.*, vol. 15, 2025, doi: 10.1038/s41598-025-15655-9.

[5] M. Fiaz *et al.*, "An explainable hybrid deep learning framework for precise skin lesion segmentation and classification," *Front. Med.*, 2025. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC12554749/

[6] M. Arshad *et al.*, "Multiclass skin lesion classification and localization from dermoscopic images," *BMC Med. Inform. Decis. Mak.*, vol. 25, 2025, doi: 10.1186/s12911-025-03051-2.

[7] S. M. M. Abohashish, H. H. Amin, and E. I. Elsedimy, "Enhanced melanoma and non-melanoma skin cancer classification using a hybrid deep learning model," *Sci. Rep.*, vol. 15, 2025, doi: 10.1038/s41598-025-08954-8.

[8] J. Vieira *et al.*, "Deep learning approaches for skin lesion detection: A benchmarking study," *Electronics*, vol. 14, p. 2785, 2025, doi: 10.3390/electronics14142785.

[9] K. Nawaz *et al.*, "Skin cancer detection using dermoscopic images with FCDS-CNN," *Sci. Rep.*, vol. 15, 2025, doi: 10.1038/s41598-025-91446-6.

[10] H. A. Owida *et al.*, "A deep learning-based dual-branch framework for automated skin lesion segmentation and classification," *Sci. Rep.*, vol. 15, 2025, doi: 10.1038/s41598-025-21783-z.

[11] S. S. Han, I. J. Moon, and W. Lim, "Ensemble learning using EfficientNet and DenseNet for improved skin lesion classification," *Comput. Methods Programs Biomed.*, vol. 250, p. 108220, 2024, doi: 10.1016/j.cmpb.2024.108220.

[12] R. Sharma, S. Gupta, and N. Patel, "Transfer learning with EfficientNet-B3 for melanoma recognition: A computationally efficient approach," *Biomed. Signal Process. Control*, vol. 92, p. 105020, 2024, doi: 10.1016/j.bspc.2024.105020.

[13] S. Chaturvedi *et al.*, "Survey and comparative analysis of CNN models for skin lesion classification (2016–2023)," *J. Biomed. Inform.*, p. 104521, 2024, (in press).

[14] G. Yang *et al.*, "Advancements in skin cancer classification: A review of modern deep learning approaches," *Multimedia Tools Appl.*, 2024, doi: 10.1007/s11042-024-19298-2.

[15] E. H. Houssein *et al.*, "An effective multiclass skin cancer classification approach based on deep convolutional neural network (DCNN)," *Cluster Comput.*, 2024, doi: 10.1007/s10586-024-04540-1.

[16] R. Sabir *et al.*, "Classification of melanoma skin cancer based on image segmentation and deep learning," *Sci. Rep.*, vol. 14, 2024, doi: 10.1038/s41598-024-75143-4.

[17] M. M. Rahman, M. R. Islam, and M. S. Hossain, "Vision transformer-based melanoma detection: Interpretability and performance comparison with CNNs," *Diagnostics*, vol. 13, p. 34, 2023, doi: 10.3390/diagnostics13010034.

[18] Y. Liu, J. Wang, and H. Chen, "Attention-guided convolutional neural networks for skin lesion classification," *BMC Med. Imaging*, vol. 23, p. 127, 2023, doi: 10.1186/s12880-023-00941-0.

[19] J. Yang *et al.*, "MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Sci. Data*, vol. 10, p. 41, 2023, doi: 10.1038/s41597-022-01721-8.

[20] P. Tschandl *et al.*, "Benchmarking deep learning models for skin lesion classification across multiple datasets," *Sci. Rep.*, vol. 12, p. 10547, 2022, doi: 10.1038/s41598-022-10547-6.

[21] J. Yang, R. Shi, and B. Ni, "MedMNIST classification decathlon: A lightweight AutoML benchmark for medical image analysis," in *Proc. IEEE 18th Int. Symp. Biomed. Imaging (ISBI)*, pp. 191–195, 2021, doi: 10.1109/ISBI48211.2021.9434062.

[22] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, pp. 6105–6114, 2019.

[23] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, p. 180161, 2018, doi: 10.1038/sdata.2018.161.